

Detection of Intracranial Hypertension Using Optic Disc Photographs Under Severe Data Imbalance

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Introduction

- Conditions such as brain tumors can cause increased pressure within the skull which leads to Intracranial hypertension (ICH)
- Initial symptoms of ICH such as headaches, nausea, and vomiting are non-specific and common
- Up to 72% of children with brain tumors are not diagnosed at the first visit for their symptoms
- A quick and non-invasive method to detect increased risk of ICH is crucial to improve clinical care of children

Proposed Solution

- Utilize fundus images to detect ICH

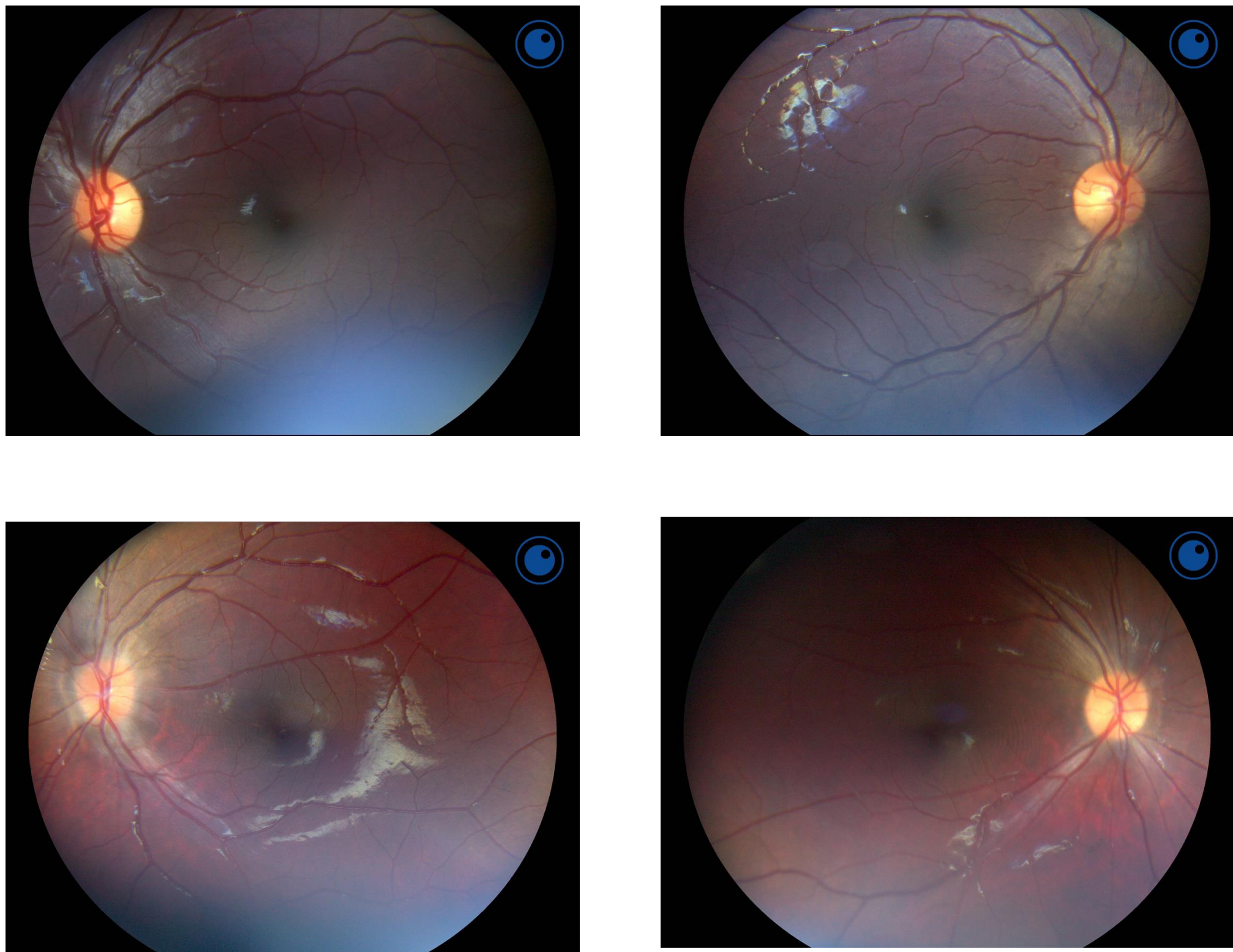


Figure 1. Can you guess which photo(s) belong to a patient with Intracranial hypertension?

- The diagnosis is difficult even for the experts
- What if we use AI tools?

Main Challenges

- **Data is Scarce**
 - There are no public datasets for ICH
 - Our curated dataset has only 458 images
- **Data is Highly Imbalanced**
 - Only ~10% samples have confirmed ICH
- **Models Overfit**
 - Difficult to convince a model not to predict “no ICH” for everything

Remedies

- **Weighted Losses:** Assign a higher weight the smaller more important class
 - Weighted Cross Entropy: $\mathcal{L} = - \sum_{c=1}^C w_c y_c \log(p_c)$
 - Focal Loss: $\mathcal{L} = -\alpha (1 - p_t)^\gamma \log(p_t)$
- **Sampling:** Equalize the samples for each class
 - Under Sampling: Reduce the size of the bigger class
 - Over Sampling: Increase the size of the smaller class
- **Foundation Models:** Use bigger models trained on more data
 - RETFound [1]: ViT-based model trained on large set of fundus images

Results

- Table 1: The effectiveness of methods utilized to combat data scarcity and imbalance. Reported values are the average (std) AUROC across 5-fold cross-validation.

	Base Model	Weighted CE	Focal Loss	Under Sampling	Over Sampling
ResNet50	0.640 (± 0.174)	0.780 (± 0.131)	0.772 (± 0.131)	0.700 (± 0.166)	0.816 (± 0.100)
ViT-L/16	0.680 (± 0.128)	0.681 (± 0.141)	0.616 (± 0.118)	0.517 (± 0.076)	0.758 (± 0.101)
RETFound	0.685 (± 0.127)	0.754 (± 0.118)	0.783 (± 0.097)	0.735 (± 0.096)	<u>0.795</u> (± 0.069)

- Surprisingly, ResNet50 has the highest performance
 - With limited data, larger models tend to overfit.
- However, ResNet50 has a much higher standard deviation
 - RETFound might a more suitable choice

Abolition Studies

- Table 1: The effect of the parameter alpha in focal loss. Reported values are the average (std) of metrics across 5-fold cross-validation.

	$\alpha=.1$	$\alpha=.2$	$\alpha=.25$	$\alpha=.3$	$\alpha=.35$
AUROC	0.669 (± 0.091)	0.773 (± 0.090)	0.782 (± 0.092)	0.783 (± 0.097)	0.781 (± 0.098)
Sensitivity	1.000 (± 0.000)	1.000 (± 0.000)	1.000 (± 0.000)	0.808 (± 0.164)	0.164 (± 0.120)
Specificity	0.000 (± 0.000)	0.000 (± 0.000)	0.005 (± 0.006)	0.608 (± 0.125)	0.988 (± 0.019)

Acknowledgement



- This work was conducted in collaboration with Dr. Melinda Chang at Children’s Hospital of Los Angeles.

References

[1] Zhou, Yukun, et al. "A foundation model for generalizable disease detection from retinal images." Nature 622.7981 (2023): 156-163.